Mauricio S. Callahan

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Education

Miami University Oxford, OH

Bachelor of Arts in Computer Science, Minor in Finance *May 2027*

Relevant Coursework: Database Systems, Introduction to AI, Web Application Programming, Data Abstractions & Structures, Algorithms, Financial Modeling in Excel

Experience

Software Engineering Intern, Payments June 2026 – August 2026

JPMorgan Chase *Columbus, OH*

- Built a Spring Boot bulk ACH cancellation API so onboarding clients can cancel all test transactions at once
- Saved about 15s per transaction by accepting client reference numbers instead of manually looking up identifiers
- Designed a PostgreSQL workflow that atomically cancels any size batch of transactions in just 2 queries
- Led an AI assisted build with GitHub Copilot, specifying each method and reviewing all generated code
- Created an AI agent that translates plain English into Splunk queries through JPMC's internal MCP server

Peer Career Coach August 2025 – Present

Miami University – Center for Career Exploration and Success *Oxford, OH*

- Conduct 1:1 resume reviews, mock interviews, and provide career development guidance to students
- Facilitate workshops and deliver 3+ presentations per semester on professional and career readiness
- Collaborate with Peer Career Coaches and career staff to improve coaching strategies and interview preparation resources

Projects

FIRE! Budgeting App (In Progress) | *Next.js, TypeScript, Java, Spring Boot, Supabase, PostgreSQL*

- Designed a PostgreSQL ledger with triggers and ordered row locks that keep totals accurate under concurrency
- Secured user data with row level security and column level permissions, verified by automated SQL tests
- Built a stateless Spring Boot backend validating Supabase auth tokens, with JUnit tests against a mock server

Riven Market Pricing Tool | *Python, Flask, React, TypeScript* March 2026

- Built a Flask backend estimating prices for player traded digital items from live auction data
- Developed a similarity based price estimator with outlier removal, age decay, and market derived stat weights
- Implemented rate limiting, 24-hour caching with fallback, validated with unit tests

Pose Detection | *Python, PyTorch, NumPy* May 2026

- Trained a custom CNN and modified ResNet18 with a 4 person team to predict 17 body joints from radar data
- Built MMVR preprocessing with Gaussian heatmap targets, reaching 0.931 PCK@5% on a 200 sample subset

Extracurricular Activities

President April 2025 – May 2026

Society of Hispanic Professional Engineers (SHPE) *Oxford, OH*

- Led 10+ networking and professional development events with corporate recruiters and engineering professionals
- Served as liaison between the chapter, National SHPE, CCES, and the engineering department

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML, CSS
Frameworks & Libraries: Spring Boot, Next.js, Flask, React
Databases: PostgreSQL, Supabase
Cloud & DevOps: Docker, Kubernetes, Jenkins, Linux
Developer Tools: CI/CD, Git, GitHub, Bitbucket, Jira, Maven, Splunk, Bruno, REST APIs
AI Tools: GitHub Copilot, MCP, Claude Code, Codex